

PRACHI DESAI

AI/ML Engineer | Data Engineer | Microsoft Certified Data & AI Professional

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience developing enterprise-grade data platforms, machine learning solutions, and cloud-native analytics applications. Strong foundation in Deep Learning, Python, Azure AI, and production data engineering, complemented by MTech research specializing in advanced neural network architectures. Experienced in designing scalable AI pipelines, predictive models, intelligent automation, and business intelligence systems across enterprise environments. Microsoft Certified Data & AI Professional with expertise spanning Machine Learning, Deep Learning, Generative AI, Azure Machine Learning, SQL, Docker, Fast API, and MLOps. Passionate about building production-ready AI systems that deliver measurable business impact.

CORE COMPETENCIES

Machine Learning • Deep Learning • Generative AI & LLMs • Retrieval-Augmented Generation (RAG) • Natural Language Processing • Computer Vision • Predictive Analytics & Forecasting • Data Engineering • MLOps & Model Deployment • Azure Cloud Architecture • Business Intelligence (Power BI) • SQL Performance Optimization • REST API Development • Solution Architecture • Stakeholder Management

TECHNICAL SKILLS

- **Programming:** Python, SQL, C++, Java
- **Machine Learning:** Scikit-learn, XGBoost, CatBoost, LightGBM, Random Forest, SVM, KNN, Naive Bayes
- **Deep Learning:** TensorFlow, PyTorch, CNN, RNN, LSTM, GRU, Transformers, Attention Mechanisms, Transfer Learning
- **Generative AI:** LLMs, OpenAI API, Azure OpenAI, LangChain, RAG, Prompt Engineering, Vector Databases, Embeddings
- **Computer Vision:** OpenCV, YOLO, Image Classification, Object Detection, OCR
- **NLP:** BERT, RoBERTa, Hugging Face Transformers, Sentence Transformers, Text Classification, NER
- **Cloud (Azure):** Azure Machine Learning, Azure AI Services, Azure Functions, Azure Storage, Azure Databricks, Azure Data Factory
- **MLOps & APIs:** Docker, MLflow, Git, GitHub, CI/CD, FastAPI, REST APIs
- **Databases:** SQL Server, PostgreSQL, MongoDB, Azure SQL
- **Visualization:** Power BI, Tableau, Plotly, Matplotlib

PROFESSIONAL EXPERIENCE

Application Developer | AI & Data Engineer

2022 – Present

Dhyey Consulting Services Pvt. Ltd.

- Designed and delivered enterprise AI and analytics solutions for global clients across industries.
- Built predictive analytics and forecasting models using Python and machine learning techniques.
- Collaborated with business stakeholders to translate requirements into AI-enabled solutions.
- Developed scalable data pipelines integrating ERP, CRM, SQL Server, Azure, and cloud platforms.
- Engineered high-performance SQL solutions processing large enterprise datasets.
- Designed optimized Power BI semantic models with advanced DAX and Power Query transformations.
- Developed REST APIs and backend services supporting production AI workflows.
- Integrated Azure cloud services into production analytics platforms.
- Improved query performance and data refresh efficiency through optimization and indexing.
- Led technical discussions, solution architecture, and mentoring of junior developers.

Key Achievements

- Successfully delivered multiple enterprise analytics and AI-enabled solutions for global clients.
- Reduced report execution time through optimized SQL and semantic modeling.
- Automated manual reporting processes, significantly improving operational efficiency.

- Led client-facing technical workshops and pre-sales demonstrations.
- Earned multiple Microsoft Data & AI certifications while in role.
- Completed advanced MTech research in Deep Learning alongside full-time work.

AI/ML PROJECTS

Intelligent Document Processing System — *OCR, NLP, Azure AI, Python*

- Built an end-to-end document intelligence pipeline using Azure AI and OCR/NLP models to extract, classify, and structure data from enterprise documents, reducing manual processing effort.

Customer Churn Prediction — *XGBoost, SHAP, Feature Engineering*

- Developed a churn prediction model with XGBoost and SHAP-based interpretability, enabling proactive retention actions on high-risk customer segments.

Predictive Sales Forecasting — *LSTM, Prophet, Time Series Forecasting*

- Implemented time series forecasting models (LSTM and Prophet) for sales planning, improving forecast accuracy and supporting data-driven inventory and revenue decisions.

AI Chatbot using LLM + RAG — *LangChain, OpenAI, Azure OpenAI, FAISS, FastAPI*

- Designed a retrieval-augmented generation chatbot with LangChain, Azure OpenAI, and a FAISS vector store, served via FastAPI for enterprise knowledge-based Q&A.

Computer Vision Inspection System — *PyTorch, YOLO, OpenCV*

- Created an automated visual inspection system using YOLO object detection and PyTorch to identify defects in real time, improving quality-control consistency.

M.TECH RESEARCH

Deep Learning-Based Intelligent Prediction Framework

- Designed a novel deep learning architecture for intelligent prediction tasks.
- Conducted extensive experimentation on real-world datasets.
- Evaluated performance against traditional machine learning algorithms.
- Performed systematic hyperparameter optimization and advanced feature engineering.
- Published academic research based on the framework.

PUBLICATIONS

- Research publication on a deep learning-based intelligent prediction framework, MTech thesis work. Full citation available on request.

CERTIFICATIONS

- Microsoft Certified: Azure Data Scientist Associate (DP-100)
- Microsoft Certified: Power BI Data Analyst Associate (PL-300)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

EDUCATION

MTech — Deep Learning / Computer Science

Parul University, Vadodara

BE — Computer Engineering

Parul University, Vadodara

Language

English | Gujarati | Hindi